

URL Shortener

How It Works — An Overview for Everyone

What is this?

This application turns long, unwieldy web addresses into short, shareable links — the same idea behind well-known services like bit.ly. You paste in a long link, the system hands you back something short and clean, and clicking that short link takes anyone straight to the original destination.

It was built as a personal project with a strong focus on doing the security fundamentals properly, even at a small scale — this document explains how it works and what protects your data along the way, without assuming any technical background.

How it works, step by step

- **1.** Paste your long web address into the box on the site and click “Shorten It!”
- **2.** The system generates a short, unique code and gives you back a new link built around it.
- **3.** You copy and share that short link however you like — email, social media, a text message.
- **4.** Anyone who clicks it is automatically sent straight to your original long link. They don't see or need to know what the destination address was ahead of time.

Nothing about this requires an account, a password, or a sign-up. Your browser is quietly given its own private identifier the first time you use the site, which is how the system recognises “your” links well enough to show you a list of them and let you delete them — without ever knowing who you actually are.

Security and privacy features

A lot of care has gone into protecting this small application the way a much larger, professionally-run service would be. Here's what's in place, explained plainly:

Your links are encrypted before they're ever stored

The system doesn't just save your original web address as plain, readable text in its database. Before anything is written to storage, it's scrambled using strong encryption, unique to that specific link. Even in the unlikely event that the underlying storage was somehow exposed, the actual addresses inside it would not be readable.

Only you can see or delete your own links

Each browser is quietly given a private identifier the first time it's used, and every link you create is tagged with it behind the scenes. This is what lets the site show you “your” links and let you delete them, without any login — and it also means a stranger can't delete a link just because they happen to know its short code.

The service can't be flooded or abused

Limits are built in on how many links can be created in a short window, both per visitor and across the whole site per hour and per month. This protects the service from being overwhelmed by automated bots or

accidental runaway scripts, and keeps it running smoothly for everyone.

A protective layer sits in front of everything

All traffic passes through Cloudflare, a company that specialises in web security and is used to protect a huge portion of the internet. It filters out obvious automated scanning and bot traffic before it ever reaches this application, and provides the secure padlock (HTTPS) connection you see in your browser.

The real server can't be reached directly

The actual backend server that does the work is deliberately locked down so it will only accept traffic that has come through that protective Cloudflare layer first. Even someone who discovered the server's real underlying address couldn't connect to it directly — the door is simply closed to anyone else.

A secret handshake confirms every request is genuine

On top of that network-level lock, the protective layer and the backend server also share a secret password of sorts, invisibly attached to every legitimate request. If that secret isn't present and correct, the request is rejected — so even a request that somehow looked otherwise valid still wouldn't get through without it.

Links don't linger forever

Shortened links are automatically cleaned up and removed after a set number of days. This keeps the system tidy and avoids old, forgotten links sitting around indefinitely.

Why there are usage limits

You may notice the service enforces a monthly and hourly cap on how many links can be created. This isn't a security measure so much as a practical one — this project runs on a free tier of cloud hosting, which comes with a fixed monthly allowance of usage before any cost applies. The limits simply keep the project comfortably within that free allowance.

In summary

This is a small, personal project — but it's been built with the same layered, defence-in-depth thinking used in much larger production systems: encryption of stored data, strict access controls on who can reach the server at all, protection against automated abuse, and a design that never requires collecting personal information from anyone who uses it.